Addendum: R^2 scans over Chris's parameters

26 September 2026

1 Method

The model is column BH of the tab *Copy of Copy of Copy of density*. It is computed with the sheet's formulas unchanged ($\mathsf{T} = \mathsf{R} + \mathsf{K}\,dt$, $dt = 1$, $G = 6.67 \times 10^{-17}$, $R = 1.3 \times 10^{23}\,\mathrm{km}$, $k = 2.18 \times 10^{-18}\,\mathrm{s}^{-1}$). The code reproduces the cached BH to 1.6×10^{-7}. The fit quality is

$$R^2 = 1 - \frac{\sum_i (y_i - \mathrm{BH}_i)^2}{\sum_i (y_i - \bar{y})^2}, \tag{1}$$

with no refit of slope or offset. Three targets are used: the 71 Migkas points at $\theta = \ell$; $A_F(\ell)$ on a 1° grid; and, as a secondary target, the sheet's own lookup-mapped Migkas column AG. The optimiser is a 400-point random screen followed by 25 Nelder–Mead starts. Bounds are $1.05 \le D/R \le 10$, $0.02 \le L/D \le 3$, $0.1 \le M/M_{\mathrm{sheet}} \le 10$, $|\mathrm{BA57}| \le 0.15$, and phase in $[0, 2\pi)$. None of the best fits lies on a bound.

Independent combinations. BH depends on D/R, L/D, BA57, BA59 and

$$\Gamma \equiv \frac{G\,M\,dt}{D^2\,k\,R} \qquad (\text{sheet: } 0.247). \tag{2}$$

G, dt, k and R enter only through Γ. Scaling k by s is identical to scaling M by $1/s$, which was checked to machine precision. So k is not scanned separately, and M stands for the gravity strength. Changing the sign of BA57 and adding π to BA59 gives the same curve; the tables quote the form with BA57 < 0.

The optional offset ℓ_0 sets $\theta = \ell - \ell_0$ and is *not* in the sheet, where $\theta = 0$ corresponds to $\ell = 0$.

2 Best fits

Case	C23	D23	B27 M (kg)	BA57	BA59 (rad)	ℓ_0	Γ	R^2_{Migkas}	$R^2_{A_F}$	$R^2_{\mathrm{sheet\,map}}$
Sheet values	5	0.18	4.44×10^{68}	−0.06	0.35	–	0.247	0.632	0.691	0.706
Best vs Migkas	2.26	0.295	6.33×10^{67}	−0.047	0.21	–	0.173	**0.759**	0.833	0.579
Best vs A_F	2.10	0.291	5.08×10^{67}	−0.046	0.16	–	0.161	0.759	**0.834**	0.573
Best vs sheet map	3.59	0.069	1.48×10^{68}	−0.060	0.83	–	0.160	0.588	0.641	**0.790**
Optional, with axis offset ℓ_0 (not in the sheet):										
Best vs Migkas	2.45	0.297	9.05×10^{67}	−0.042	0.45	13.0°	0.210	**0.897**	0.981	–
Best vs A_F	2.32	0.295	7.69×10^{67}	−0.041	0.41	12.9°	0.199	0.896	**0.982**	–

For reference, A_F itself scores $R^2 = 0.915$ against the 71 points.

3 Profiles

Figure 1 shows R^2 against the Migkas points ($\theta = \ell$) for each parameter. One curve holds the other parameters at the sheet values; the other re-optimises them.

Cell	Sheet	Best	Range with $R^2 \ge R^2_{\max} - 0.01$	Others at sheet values: best value, R^2	
C23 D/R	5	2.26	1.97–3.12	5.18	0.636
D23 L/D	0.18	0.295	0.26–0.30	0.26	0.680
B27 M (kg)	4.44×10^{68}	6.3×10^{67}	$\le 4.4 \times 10^{67}$ (bound) – 1.6×10^{68}	4.2×10^{68}	0.637
BA57	−0.06	−0.047	−0.053 – −0.038	−0.053	0.638
BA59 (rad)	0.35	0.21	−0.1 – 0.47	0.63	0.656
ℓ_0 (optional)	0	13°	10–15°	10°	0.715

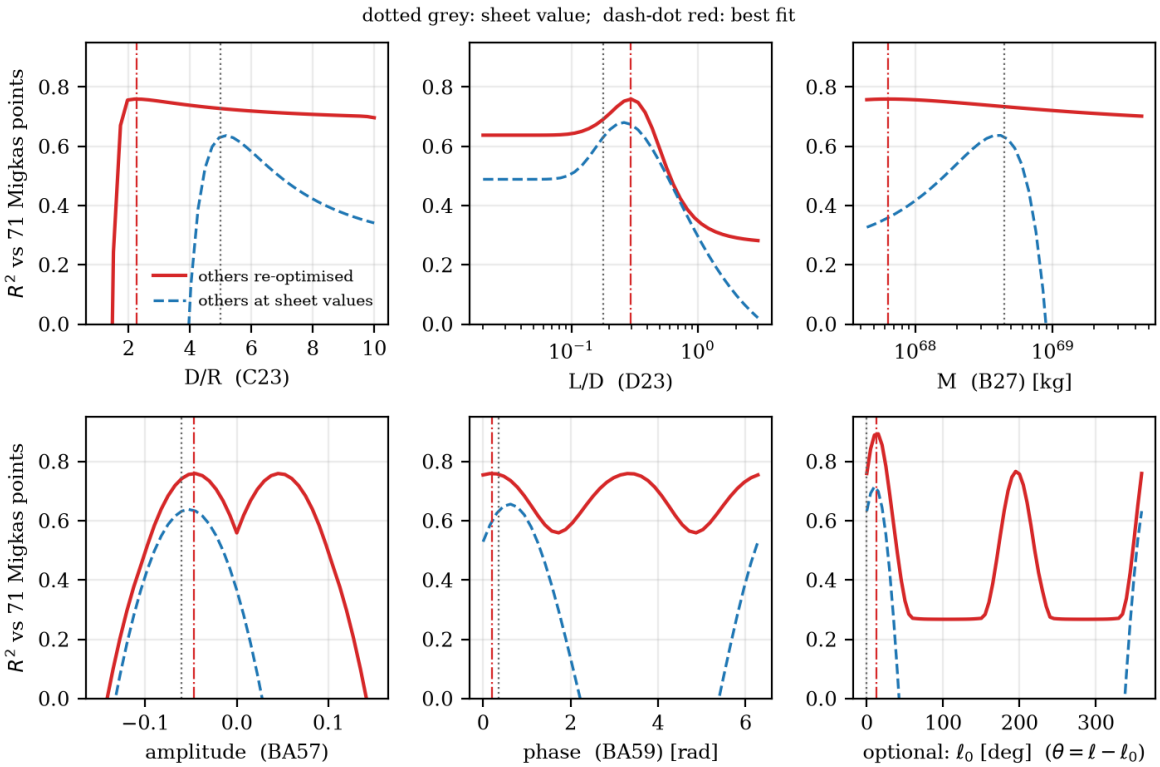


Figure 1: R^2 profiles against the Migkas points. Solid: others re-optimised. Dashed: others at sheet values. The last panel is the optional ℓ_0.

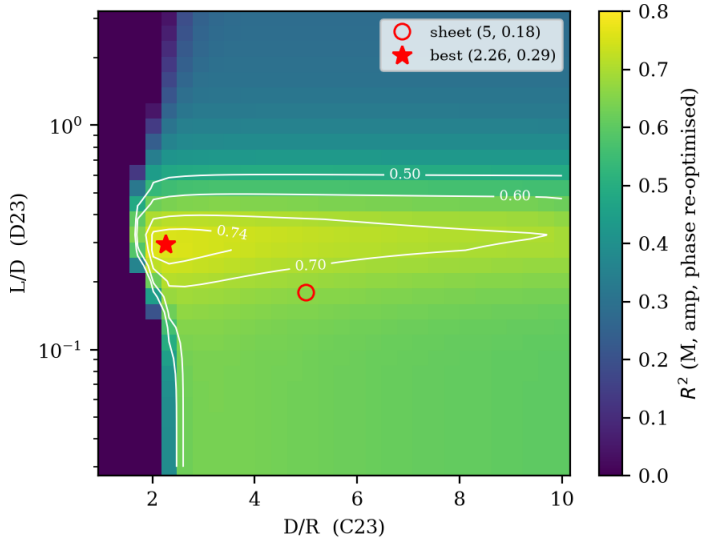


Figure 2: R^2 over D/R and L/D, with M, BA57 and BA59 re-optimised ($\theta = \ell$).

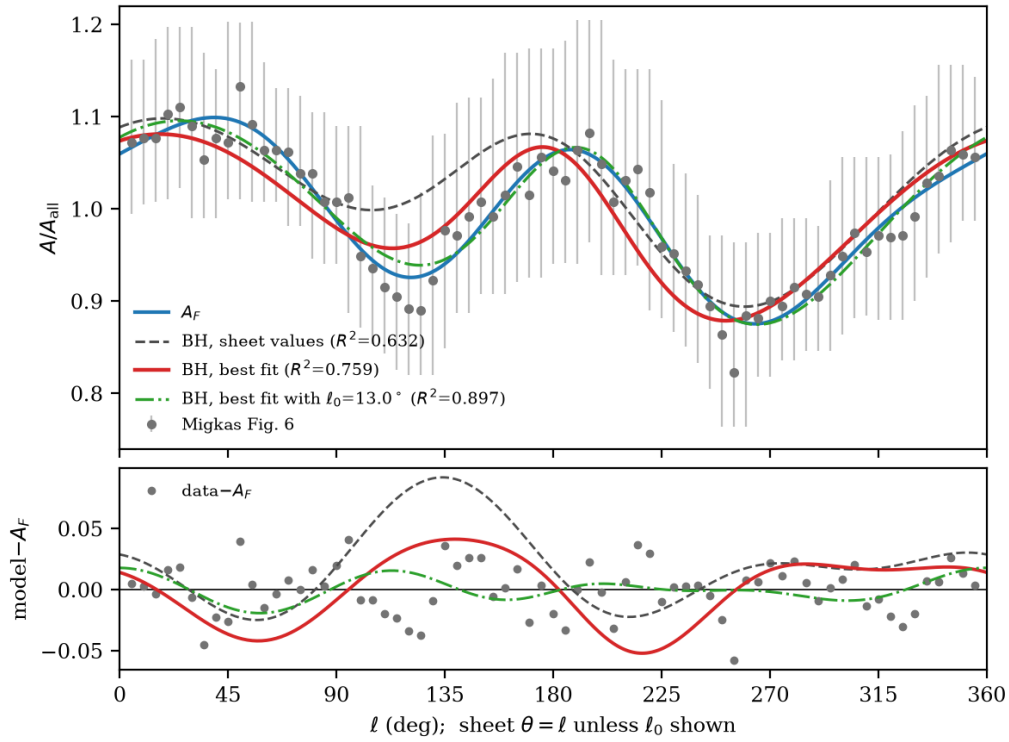


Figure 3: BH at the sheet values, at the best fit, and at the best fit with ℓ_0, compared with A_F and the Migkas points.

4 Notes

1. D/R and M are strongly degenerate. Along the D/R profile the best M rises roughly as $(D/R)^3$ and R^2 falls only slowly, from 0.759 to 0.70 at $D/R = 10$. Above $D/R \approx 9.8$ this re-optimised M reaches the upper bound. Below $D/R \approx 1.8$ the fit collapses.
2. L/D is the best-determined parameter (0.26–0.30). The sheet value 0.18 lies outside this range.
3. Without the sine term (BA57 = 0, geometry re-optimised) the best R^2 is 0.559, so the sine term adds 0.20.
4. The largest single gain is the optional $\ell_0 \approx 13°$, which raises R^2 from 0.759 to 0.897 against the points and to 0.98 against A_F. The profile has a second, lower peak near $\ell_0 \approx 195°$ ($R^2 \approx 0.77$).
5. Fitting the sheet's own lookup mapping favours different geometry ($L/D = 0.07$). Its best fits score only 0.59 against the points at $\theta = \ell$.

Files in `papers/spreadsheet_model/`: `param_scan_core.py`, `param_scan.py`, `param_scan_profiles.py`, `param_scan_plots.py`, `param_scan_results.json`, `param_scan_profiles.json`, `param_scan_map.npz`.